

OS-8

子二甲 U13157005 吳翊弘

FreeRTOSMP

EXPLORER

- FREERTOSMP
 - .vscode
 - build
 - lib
 - port
 - src
 - Agent.cpp
 - Agent.h
 - BlinkAgent.cpp
 - BlinkAgent.h
 - CMakeLists.txt
 - CounterAgent.cpp
 - CounterAgent.h
 - main.cpp 3
 - .gitignore
 - CMakeLists.txt
 - FreeRTOS_Kernel_import.cmake
 - pico_sdk_import.cmake

main.cpp 3

```
1  /**
10
11  * U13157005 吳翊弘
12
13  #include "pico/stdlib.h"
14
15  #include "FreeRTOS.h"
16  #include "task.h"
17  #include <stdio.h>
18  #include <math.h>
19  #include "hardware/structs/sio.h"
20
21  #include "BlinkAgent.h"
22  #include "CounterAgent.h"
23
24
25  //Standard Task priority
26  #define TASK_PRIORITY      ( tskIDLE_PRIORITY + 1UL )
27
28  //LED PAD to use
29  #define LED_PAD            8
```

PROBLEMS 3 OUTPUT DEBUG CONSOLE TERMINAL PORTS SERIAL MONITOR

+ Open an additional monitor

Monitor Mode Serial View Mode Text

Port COM1 - 通訊連接埠 (COM1) Baud rate 115200 Line ending None

Start Monitoring

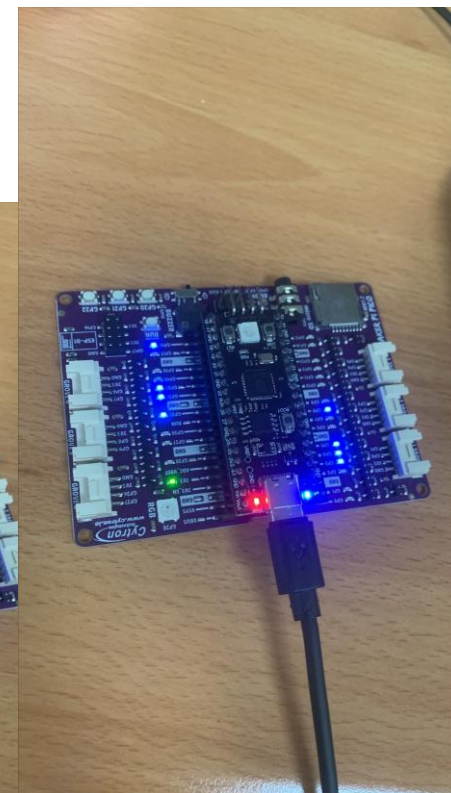
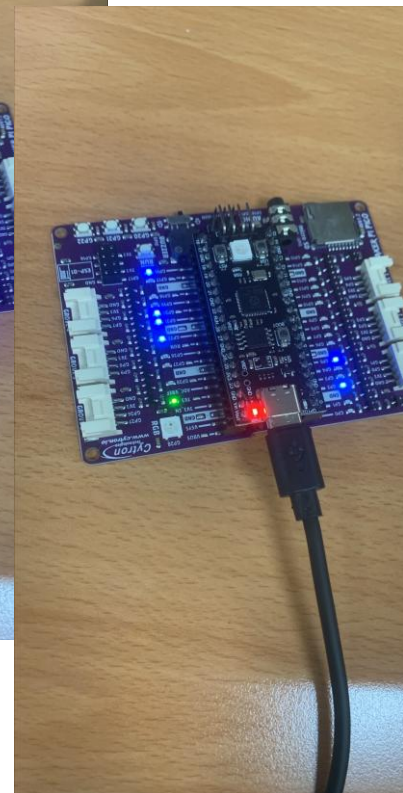
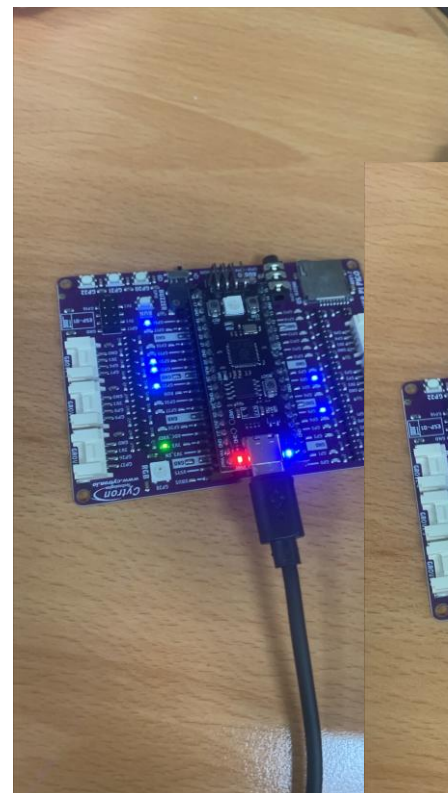
Counter on Core 1
Blink 0 on Core 0
Blink 1 on Core 1
Counter on Core 1
Counter on Core 1
Blink 0 on Core 0
Blink 1 on Core 1
Number of tasks 7

Task: 1	cPri:1	bPri:1	hw:302	MainThread
Task: 3	cPri:0	bPri:0	hw:233	IDLE1
Task: 2	cPri:0	bPri:0	hw:215	IDLE0
Task: 7	cPri:1	bPri:1	hw:0	Counter
Task: 5	cPri:1	bPri:1	hw:0	Blink 0
Task: 537013748	cPri:1	bPri:537018228	hw:0	Blink 1
Task: 4	cPri:31	bPri:31	hw:967	Tmr Svc

Counter on Core 1
HEAP avl: 126576, blocks 1, alloc: 21, free: 12
Main on Core 0

Type in a message to send to the serial port. Send Ctrl + C

Ln 13, Col 23 Tab Size: 4 UTF-8 LF C++ Compile Run Pico SDK: 2.2.0 Board: pico Pico



READ ME

透過 **Agent** 架構將不同功能分成獨立任務執行；系統啟動後會有兩顆 **LED** 以固定週期持續閃爍，同時四顆 **LED** 以二進位方式顯示由主任務定期產生的隨機數值，因此會呈現規律變化的計數效果；整體運行中各任務分別分配到不同核心並透過 **Queue** 溝通，展示了多核心調度、任務管理與 **GPIO** 控制的整合應用。